COMSATS University Islamabad (CUI)

Project Proposal

for

LOCUS

Version 1.0

By

Muhammad Omair Tahir **CIIT/SP23-BCS-059/ISB**

Taqdees Zahra **CIIT/SP23-BCS-096/ISB**

Supervisor

Mumammad Rashid Mukhtar

Bachelor of Science in Computer Science (2023-2027)

Table of Contents

Project Category: (Select all the major domains of proposed project**)**

□ A-Desktop Application/Information System □ B-Web Application/Web Application based Information System □ C-Problem Solving and Artificial Intelligence □ D-Simulation and Modeling □ E-Smartphone Application
□ F-Smartphone Game □ G-Networks □ H-Image Processing
□ Other (specify category) ________________

Abstract

LOCUS (AI-Powered Wearable Memory & Cognitive Assistant) is being created to address the need for personalized cognitive support systems that assist individuals with memory impairments, cognitive challenges, and common forgetfulness through continuous context-aware monitoring. Currently those experiencing memory problems, seniors and neurotypical people facing forgetfulness rely on approaches such, as basic reminder apps, manual caregiver supervision or static medication management tools that lack real-time insights or adaptive behavior modifications. Existing systems do not possess features for anomaly detection, multi-modal memory access or tailored cognitive interventions designed for behavioral patterns. LOCUS primarily addresses problems such as solutions' inability to understand user routines identify shifts indicating health concerns and provide seamless incorporation of memory assistance combined with caregiver support, for older adults while also serving busy professionals and students who struggle with daily organization. A major gap in technologies is the absence of a single AI-driven platform that integrates pattern recognition, multi-modal memory access, social interaction support and intelligent behavior modeling into one wearable device suitable for elderly people needing substantial care as well as everyday users dealing with common forgetfulness. LOCUS aims to create a system that autonomously adjusts to user routines and provides proactive alerts, for atypical behaviors. The project aims to create a modal memory retrieval system that allows users to access their memories using visual, auditory, and time-based queries. Furthermore, LOCUS will incorporate assistance features that support individuals experiencing memory challenges or everyday forgetfulness, and routine management tools. This project will significantly enhance the quality of life and independence for individuals with cognitive challenges while improving daily organization for users, reducing caregiver burden through intelligent automation, real-time monitoring, and data-driven insights.

1. Introduction

This project proposal is a plan of development of the LOCUS (AI-Powered Wearable Memory and Cognitive Assistant) an assistive device that is directed to the support of people with cognitive impairments and to ordinary users willing to have a better memory support. The paper will outline the system structure, system specifications, implementation plan, and expected outcomes of the suggested technology. LOCUS makes use of machine learning and multi-modal sensing, as well as intelligence, to offer customized context-specific cognitive support. The problems associated with cognition affect millions of people all over the world between memory lapses that come with old age and forgetfulness experienced by students, professionals, and active individuals. The main feature of the current assistive tools is the reliance on reminders and tracking without involving adaptive intelligence or contextual awareness. Intelligence, computer vision and natural language processing have advanced to the point that it is possible to build systems to act as cognitive aids to users, of any age group. LOCUS also can transform into a reminder device into a proactive helper, by learning what is happening to the user, detecting abnormal behavior, and seeking immediate context-sensitive assistance. The platform includes routine recognition, multi-modal memory retrieval, social interaction support, protection monitoring and medication-related tasks. The controls of all features are done through a single dashboard which customizes its capabilities and authorization based on user needs and credentials. In the final analysis, the purpose of this project is to increase user autonomy, quality of life, cognitive load, and illustrate the practical use of AI-oriented wearable technology.

2. Problem Statement

Across the globe millions of people face issues with cognition and memory leading to an increasing need for cognitive assistance technologies. Present assistive options are disjointed, requiring users to juggle apps for reminders about medication, tracking whereabouts, memory support and communication with caregivers, which elevates cognitive strain and lowers usage rates. Existing GPS-based devices provide reactive location information and do not include predictive or preventive functions, in cases of unusual behavior. Worldwide 40–50% of patients do not follow their medication regimens properly resulting in healthcare expenses exceeding $100 billion. Most reminder systems do not include smart confirmation, escalation, or caregiver collaboration. Numerous people spend time looking for lost daily items while current tracking devices do not accommodate non-electronic belongings. Behavioral shifts, like withdrawal and changed routines are signs of cognitive deterioration but ongoing behavior monitoring with prompt notifications is still mostly inaccessible. Memory aid devices depend on fixed notes or images without question-driven access restricting their effectiveness, for people facing memory challenges. Care providers undergo physical and emotional pressure while missing immediate awareness of everyday habits, medication compliance and worrisome behavior trends. Privacy issues additionally obstruct usage since numerous systems do not provide data management or privacy protections focused on the user. These gaps highlight the need for an integrated, AI-powered, privacy-conscious cognitive assistant like LOCUS that delivers proactive, adaptive, and unified cognitive support through a single wearable platform.

3. Problem Solution/Objectives of the Proposed System

LOCUS is going to overcome such difficulties by creating AI-based wearable cognitive assistant that will combine different assistive features into one smart platform. This solution makes use of machine learning techniques to understand the habits and behavior patterns of every user and removes the necessity, to be configured and instead offers individual users cognitive assistance that modifications to the unique requirements of that individual. The combination of pattern recognition and anomaly detection features make LOCUS turn technology into an active framework instead of a reactionary one in that it detects abnormal behavior and notifies users of their actions before they get into trouble instead of responding to them after they get in trouble. The multi-modal memory recall system overcomes a deficiency, in technologies, by allowing users to retrieve memories using natural language queries which unite visual, auditory, temporal, and contextual data to assist individuals to re-experience key daily experiences and convey the whereabouts of lost objects. LOCUS addresses issues of non-compliance with medication through utilizing verification and better alert systems and effective integration with caregiver dashboards effectively lowering the 40-50% non-adherence rate using technology-enabled monitoring. The comprehensive location safety solution is a solution that deals with wandering issues, including offering real-time tracking, geofence notifications, an emergency I'm Lost feature with a panic button and guided home directions that is specifically focused on people to wandering tendencies. Using smart behavioral modeling, LOCUS provides memory support and overall cognitive needs with customization; that is, it no longer uses a one-size-fits-all model to give interventions that do not address individual needs of users.

3.1 Objectives

BO-1: Achieve 80-90% accuracy in automated learning and the identification of behavior patterns.

BO-2: Decrease the occurrence of medication non-adherence by 60% using clever checks, increasing reminders, and integrating caregivers.

BO-3: Reduce the time spent locating lost items by 70% for users by utilizing multi-modal memory retrieval and visual search functions.

BO-4: Cut wandering-related accidents by 50% among the elderly users by detecting anomalies proactively and real-time monitoring of locations.

BO-5: Reduce critical behavior changes response time by 80% using real-time dashboard notifications and automatic communications to elderly users.

BO-6: Increase user independence scores by 35% among the elderly users as indicated by the lowering of daily caregiver intervention needs.

BO-7: Achieve 85% user satisfaction rating for social interaction support through face recognition and contextual prompting features across all user groups.

BO-8: Ensure 100% compliance with privacy regulations through end-to-end encryption, automatic data expiry, and granular user controls.

4. Related System Analysis/Literature Review

Considerable advancements have occurred in technology through wearable camera devices voice-activated AI assistants and specialized memory aid tools each addressing unique aspects of cognitive support but lacking complete integration. FMT (Fiducial Marker Tracker) developed by Li et al. (2019) Was a pioneer in object tracking with a neck-worn camera that captured video clips when visual markers, in positions were identified. This system demonstrated that automated object tracking could be achieved but required affixing markers to objects limiting its practicality in situations. Because it depended on markers users had to decide in advance which items to monitor, obstructing recall of unmarked objects.

GO-Finder, presented by Sato (2021) advanced the field by eliminating the necessity for registration through the identification and grouping of objects, in real-time video streams. This approach addressed the problem of wasted time with research indicating daily durations spent searching for lost items. However, GO-Finder focused on visual object retrieval and did not incorporate the learning of behavioral routines medication reminders or caregiver participation. Additionally, the system lacked multimodal search capabilities that combine auditory and temporal data for a comprehensive memory reconstruction.

The first study to be mentioned, SenseCam by Hodges et al. (2006) revealed the importance of the first-person photo capture to improve memory recall and lifelogging features. This new technology demonstrated how wearable cameras could be used continuously to capture the occurrences which could be reviewed later. Although it was improved SenseCam was essentially an instrument of recording without intelligent search capabilities, real-time notifications or intelligent assistance. This put the user at the disadvantage of having to sift through thousands of images, which puts a cognitive load, which these people already had in terms of memory problems.

MemPal, created by Maniar et al. (2024) was a breakthrough because it introduced the integration of cameras with Vision-Language Models to offer multimodal AI voice-operated memory assistance. The system created the text journals based on the camera images that could be read by the users with the help of natural language voice commands that were effective to prove the potential of LLMs to comprehend the context and facilitate recall memory. MemPal confirmed such design principles as the preference of the user to interfaces instead of just visual interfaces and the possibility of retrieving the objects without prior registration with the help of VLM technology. The prototype featured an exception by having a camera on the neck. It was superior in object location tasks that set the foundation, to the multimodal assistant model that directly impacts on LOCUSs design approach.

A recent study conducted by O'Brien et al. (2022) On Voice-Controlled Intelligent Personal Assistants like Amazon Alexa and Google Home indicated that seniors also knew how to use them, and they had such technology available in their lives. These VIPAs offered companionship, home reminders and home management to help decrease the isolation among the users. Nevertheless, general-purpose voice assistants do not include access to cognitive support, including visual memory search, wandering detection, medication verification or sensor integration, as a continuous monitoring. They are too general in their approach, which do not satisfy the needs of assistance.

Tsiourti et al. (2014) Carried out qualitative field research on companions to determine key design considerations in acceptance of technology by the older adults. Their study emphasized on the importance of user interfaces, ease of installing, privacy, and social acceptance in developing technologies. The results revealed that users like those systems that are said to support, followed by control with emphasis on keeping user control. These lessons guide the design plan of LOCUS. Cognitive aid Reflect user research, rather than a working system, which leads to gaps in transforming these principles into a fully-fledged cognitive aid.

The current systems jointly show the viability of the components, including the object tracking, voice interaction and passive observation but show significant gap in the integrated proactive cognitive assistance. None of the existing solutions combines learning, multi-modes of memory retrieval, intelligent analysis of behavior, medication, location safety and caregiver coordination into one platform. The main mode of operation of these systems is through dependency on the user to either demand or seek information as opposed to actively identifying anomalies and averting incidents. LOCUS can overcome these constraints by integrating what has been learned amid current research and launching intelligent pattern recognition, in-depth caregiver dashboard, and specialized support of various cognitive requirements in a connected wearable ecosystem.

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
FMT (Fiducial Marker Tracker) - Li et al., 2019	Requires manual placement of visual markers on objects for tracking, limiting spontaneous memory recall. Only tracks pre-registered items, cannot help finding unmarked objects. Lacks behavioral learning and caregiver integration.	LOCUS implements registration-free object tracking through continuous visual monitoring combined with AI-powered scene understanding. Automatically identifies and tracks all objects without markers. Integrates routine learning to predict which items user typically carries.
GO-Finder - Sato, 2021	Focuses solely on visual object retrieval without multi-modal context. No behavioral pattern learning or proactive alerts. Lacks medication management, location safety, and caregiver features. Registration-free but limited to hand-held object discovery only.	LOCUS combines visual tracking with audio context, temporal data, and location information for comprehensive memory reconstruction. Adds intelligent routine learning, medication reminders, wandering detection, and real-time caregiver dashboard for holistic cognitive support.
SenseCam - Hodges et al., 2006	The device is a passive recorder. Users will need to manually check each image, perhaps thousands of them. Users cannot search images intelligently. Therefore, the process is burdensome to the user, as they must compare, search through, and find relevant images from their vast database of images, as it does not have a proactive ability to assist users or provide aid.	LOCUS provides AI-powered natural language search enabling queries like "where did I see my keys yesterday." Implements proactive anomaly detection and real-time alerts. Automatically organizes memories with intelligent indexing, eliminating manual review burden.
MemPal - Maniar et al., 2024	It is excellent at VLM-powered object retrieval and natural language queries but does not provide routine learning and anomaly detection. Moreover, it makes use neck-worn camera form factor which is less discreet than camera. No medication management, location safety features, or real-time monitoring. Missing condition-specific behavioral modeling for diverse cognitive impairments. No caregiver integration or emergency response capabilities.	LOCUS builds directly upon MemPal's validated multimodal paradigm by integrating VLM-powered memory assistance into camera for enhanced discretion and always-ready accessibility. Extends core functionality with intelligent pattern recognition for behavioral anomalies and proactive alerts. Adds comprehensive medication adherence system with verification, wandering detection with GPS tracking, and intelligent support modules for memory assistance and general cognitive challenges. Implements unified caregiver dashboard with real-time monitoring and emergency

		response features for holistic cognitive support beyond object retrieval.
Voice-Controlled Personal Assistants (Alexa, Google Home) - O'Brien et al., 2022	General-purpose assistants lacking specialized cognitive support features. No visual memory search or wearable sensor integration. Cannot detect wandering, verify medication, or provide condition-specific interventions. One-size-fits-all approach unsuitable for diverse cognitive needs.	LOCUS combines voice interface with wearable camera, sensors, and GPS for continuous contextual awareness. Provides specialized cognitive support including visual memory search, medication verification, wandering detection, and tailored interventions for memory assistance and general cognitive challenges.
Virtual Assistive Companions - Tsiourti et al., 2014	Represents user research and design principles rather than functional system implementation. Identified requirements for intuitive interfaces and privacy but did not deliver integrated cognitive assistance platform. No technical solution for memory recall or safety monitoring.	LOCUS translates research insights into comprehensive functional system with intuitive voice/visual interface, minimal setup through automatic learning, granular privacy controls, and socially acceptable wearable form factor. Implements "guide rather than direct" philosophy through subtle prompts and user autonomy preservation.

5. Vision Statement

For individuals facing challenges like memory impairments and neurotypical people who sometimes forget, well as their caregivers who deal with fragmented assistive technologies, missed medication routines, lost items, risks of wandering difficulties in social interaction and ongoing concerns about overlooking essential daily tasks while caregivers experience fatigue from constant supervision needs **the** LOCUS (AI-Powered Wearable Memory & Cognitive Assistant**) is** an advanced wearable cognitive aid paired with companion mobile and web applications **that** independently learns individual routines actively detects behavioral anomalies enables natural language memory inquiries across visual and auditory contexts, verifies medication adherence prevents wandering incidents through real-time location monitoring supports social engagement with discreet prompts and provides caregivers with comprehensive dashboards, for remote monitoring and timely intervention. **Unlike** currently installed fragmented reminder apps, basic GPS trackers, non-portable memory alarms, manual memory journals, general-purpose voice assistants such as Alexa and other older wearable memory systems like MemPal and GO-Finder that only handle single points of the cognitive support system and do not provide proactive prevention, **our product** serves as the first complete, full-fledged, and thoroughly integrated cognitive assistant integrating pattern recognition, multi-modal memory recall, intelligent behavioral modeling, and caregiver collaboration within a discreet, always ready camera, transforming cognitive assistance from reactive reminders to proactive prevention while preserving user independence, dignity, and privacy through intelligent automation that learns, adapts, and intervenes only when necessary.

6. Scope

The scope of LOCUS involves developing a support system consisting of a wearable camera, mobile applications, a web and app dashboard and cloud-based AI services working together to provide comprehensive memory and cognitive support. This project will implement a learning engine that continuously monitors user activities through wearable camera data automatically identifying daily routines such, as medication intake, departure routines exercise schedules and object handling patterns without requiring the user to input information manually. A multi-modal system to retrieve memories is proposed to be designed that allows the users to retrieve their personal memory database by using natural language voice recognition. This platform will use a collection of visual objects recognition, audio context recognition, filtering, and location data to answer queries such as: where am I leaving my keys or what was my conversation with John, about. In the system, there will be an anomaly detector that will compare activities to known patterns and send contextual alerts to the user in the event of irregularities, including leaving home without crucial items, forgetting medication, spending excessive time in bed, or interrupting daily routines. The overall medication adherence solution will offer verification using cameras, the computerized tracking of medication events smart escalating messages that develop to prompts and develop into critical alarms and alerts to the caregivers, when a dose is missed. The location safety function will offer GPS tracking, an emergency panic button and step-by-step voice navigation to help users navigate safely back home. Face recognition and social interaction support will enable the system to identify people and provide name reminders to maintain a relationship database with contextual information, about everyone. A unified caregiver dashboard accessible, on both web and mobile platforms will provide real-time activity monitoring, medication compliance tracking, behavioral alert notifications, location history viewing and remote assistance capabilities like sending reminders and verifying the user’s condition. Privacy and security features will include end-to-end encryption for all data transmission and storage face blurring in photos taken to protect bystander privacy, data retention options with automatic deletion comprehensive privacy settings allowing users to stop recording or exclude locations and multi-factor authentication for caregiver access. The system will offer training games and activities accessible, through the app, including spatial memory challenges, pattern recognition tasks, face- name matching exercises and an adaptive difficulty system that modifies based on user performance. The project scope explicitly includes developing machine learning models aimed at pattern recognition object detection face identification, anomaly detection, voice query handling and conversational interaction. A cloud- based system will be set up to handle video processing maintain the unified memory database perform AI calculations and synchronize data among multiple user devices and caregiver systems. The scope boundaries explicitly exclude medical diagnosis capabilities, prescription management or medication dispensing advice, emergency medical response services, and direct integration with emergency services, as LOCUS is designed as an assistive tool focusing on memory assistance and general cognitive support rather than a medical device requiring regulatory approval.

7. Modules

7.1 Camera Application Modules

7.1.1 Module 1: Intelligent Routine Learning & Behavioral Analysis

FE-1: Continuously monitor user activities through wearable camera to identify daily patterns of users.

FE-2: Learns medication times, work schedules and object handling behaviors.

FE-3: Build personalized behavioral profiles based on learned data.

FE-4: Patterns of activities like social activities.

FE-5: Create contextual notifications of exiting the house without necessary items.

FE-6: Identify skipped medication times.

7.1.2 Module 2: Multi-Modal Memory Capture & Processing

FE-1: Record video and audio streams of camera and microphone.

FE-2: Visualize data based on computer vision to identify objects and understand the scene.

FE-3: Extract audio context and convert speech to text for searchable memory database.

FE-4: Save learned patterns to local device memory and back up and analyze on the cloud.

FE-5: Turn privacy on to stop the recording temporarily.

FE-6: Enables user to stop recording automatically based on location to exclude sensitive areas.

7.1.3 Module 3: Face Recognition & Social Interaction Support

FE-1: Real time face detection in video stream.

FE-2: Compare relationship database in cloud with personal and professional contacts found on cloud.

FE-3: Show contextual information of the past interactions.

FE-4: Store new face data with user confirmation for building relationship database.

FE-5: Display important reminders about people.

7.1.4 Module 4: Location Safety & Emergency Response

FE-1: Continuously track user location using GPS and transmit data to cloud servers.

FE-2: Trigger emergency "I'm Lost" mode through voice command or panic button activation for users.

FE-3: Provide turn-by-turn voice-guided navigation to help elderly users return home.

FE-4: Activate continuous location sharing with caregivers during emergency situations for users.

FE-5: Provide two-way voice communication with designated caregivers during lost episodes.

7.2 Mobile Application Modules (iOS & Android)

7.2.1 Module 5: Activity Summary & Monitoring Services

FE-1: Display daily activity summary with key metrics (total interactions, people met).

FE-2: Show medication adherence status for users for completed and missed doses.

FE-3: Depict behavioral pattern insights with trend analysis over time.

FE-4: Provide quick access to recent memory searches and saved memories.

FE-5: Monitor storage usage and available space on device and cloud.

7.2.2 Module 6: Memory Search & Retrieval System

FE-1: Permit text search of personal memory database using natural language text query.

FE-2: Assist with voice input for memory searches.

FE-3: Search results can be filtered by date, people, or object categories

FE-4: Show search results with stills, date, and location.

FE-5: Combine visual, audio, and temporal to provide comprehensive search results.

FE-6: Timeline visualization of daily activities.

7.2.3 Module 7: Medication Management & Health Tracking

FE-1: View complete medication schedule of user with dosage information and timing.

FE-2: Get notifications about the next medication time.

FE-3: Mark the doses manually when the visual verification is not possible.

FE-4: Configure adjustable snooze medication reminders.

FE-5: View notifications of missed doses to caregivers.

7.3 Dashboard Modules

7.3.1 Module 8: Caregiver Dashboard (Elderly User Support)

FE-1: View live activity feed with the status, location, and last activity of the elderly user.

FE-2: Show live location on interactive map.

FE-3: Access emergency location information when "I'm Lost" mode is activated.

FE-4: Send personalized notifications to the mobile application of the user.

FE-5: Request status check from user with quick response options.

FE-6: Create and manage multiple caregiver accounts with role-based access permissions.

FE-7: Set notification preferences for each caregiver.

FE-8: Manage caregiver profile information and contact details.

FE-9: Download current location data with timestamp and coordinates.

FE-10: Send email notifications to caregivers based on preferences.

FE-11: Deliver SMS alerts for critical events.

FE-12: Implement escalating alert protocols with configurable timing.

FE-13: Track notification delivery status and retry failed deliveries.

7.3.2 Module 9: User Dashboard

FE-1: Display individual daily summary of activities, interactions, and key captured events.

FE-2: Show to-do and routine tasks progress such as completed, pending, delayed and rescheduled tasks.

FE-3: Display face recognition information such as summaries of people encountered recently.

FE-4: Display medication schedule and compliance such as taken, missed, and scheduled doses.

FE-5: Show alert and notification history including medication reminders.

FE-6: Enable users to confirm alerts, dismiss, or snooze alerts on the dashboard.

FE-7: Provide one-tap access to key actions including Privacy Mode and Monitoring Disabled.

FE-8: Display recent location history (if location access is enabled).

FE-9: Display storage usage for local device and cloud memory with cleanup suggestions.

FE-10: Manage notification queue for all alerts and reminders.

FE-11: Handle push notification delivery to mobile devices.

FE-12: Support notification scheduling for future reminders.

8. System Limitations/Constraints

LI-1: The camera operates through an external USB power bank. With recording (720p/1080p and Wi-Fi enabled) the duration ranges around 6–10 hours based on the power banks capacity (10,000–30,000 mAh). Recording must be manually halted by the user or caregiver to replace or recharge the power bank, fully automated 24/7 functioning is not achievable.

LI-2: The camera lacks a screen AR features, a speaker and any private audio output. Every piece of real-time information (such, as name reminders, navigation cues, medication confirmations, conversation tips, alerts) is provided via the companion smartphone’s display and speaker. In settings or if the phone is kept in a pocket or bag the user might overlook or fail to hear important notifications.

LI-3: Position tracking, step-by-step navigation, emergency "I’m Lost" functionality and communication depend completely on the smartphone’s GPS and cellular/Wi-Fi connection. These capabilities cease to function if the phone’s battery runs out the signal is lost or the device is not, with the user.

LI-4: Face recognition and social interaction assistance cannot offer immediate name prompts due to the absence of a private method to transmit confidential audio signals. Users need to look at the smartphone display (or receive alerts through vibration or loudspeaker) upon recognizing a known individual, which could draw attention or be inconvenient.

LI-5: Ongoing video capture produces 6–12 GB every hour. Insufficient SD-card and smartphone memory will require the user to halt recording if Wi-Fi's not accessible, for extended durations.

LI-6: Real-time AI functionalities (face recognition notifications, anomaly detection, cloud memory retrieval) depend on internet access through the smartphone. In regions, without connectivity these features are postponed until the internet is restored.

LI-7: The camera attached to the chest appears bulky and is less subtle compared to smart glasses, which may lower its long-term acceptance among users, in social environments.

LI-8: Voice instructions and natural language inquiries are handled exclusively by the smartphone. High ambient noise, speech impairments or the phone being, inside a pocket greatly diminish the accuracy of recognition.

LI-9: This system is not an approved device and is unable to diagnose illnesses provide medication or substitute for professional medical treatment.

LI-11: Ongoing audio recording might breach privacy regulations in some jurisdictions (such, as areas requiring two-party consent). Users need to stop recording in private settings.

LI-12: For the senior-year project sophisticated elements (precise pill identification, for every medication continuous learning model, complete EHR integration) are developed solely as working prototypes.

9. Data Gathering Approach

The needs for the LOCUS system were collected using a combination of interviews, questionnaires, and field observations. We conducted interviews with 10 individuals (5 elderly, 5 normal users) to explore daily difficulties such, as memory issues, medication management and social interaction. A survey involving 25 users and caregivers examined how often memory lapses occurred, favored methods of feedback delivery and issues related to privacy. Additionally, two users trialed a phone camera prototype during activities to assess attachment methods, comfort levels and the feasibility of recording. This combined approach ensured the functional requirements reflect real user needs, hardware constraints, and caregiver expectations.

10. Project Contribution

The LOCUS system introduces several original technical and conceptual contributions that meaningfully advance the field of wearable cognitive assistance beyond existing research prototypes and commercial solutions:

- **First integrated cognitive assistant built entirely on a consumer action camera with external power bank**: Unlike prior works (SenseCam, MemPal, GO-Finder) that used custom neck-worn cameras or assumed future smart-glasses hardware, LOCUS demonstrates that a practical, always-on, first-person memory augmentation system can be realized today using inexpensive, off-the-shelf, widely available hardware, dramatically lowering the barrier to real-world deployment.
- **Full-stack proactive cognitive support in a single wearable ecosystem without requiring AR glasses**: LOCUS is the first system to combine automatic routine learning, real-time anomaly detection, multi-modal memory recall, medication verification, wandering prevention, and caregiver dashboard within a single platform while deliberately avoiding dependence on heads-up displays or private audio channels, proving that comprehensive cognitive assistance is achievable with only a smartphone for feedback.
- **Novel anomaly-triggered "memory-on-demand" architecture**: Instead of forcing users to review hours of footage (SenseCam) or wait for post-hoc queries (MemPal), LOCUS continuously learns individual behavioral baselines and proactively surfaces relevant past video/audio clips the moment a deviation is detected (e.g., "You are leaving without your wallet – here is the last place you had it 20 minutes ago"), significantly reducing cognitive load during critical moments.
- **Zero-manual-configuration routine learning tuned for camera constraints**: A lightweight on-device + cloud hybrid pipeline that autonomously discovers medication times, departure checklists, frequently carried objects, and daily patterns from raw camera video within the first 7–14 days, eliminating the setup burden that prevents adoption in existing assistive tools.
- **Context-aware medication verification using deliberate user gesture**: A new interaction pattern where the user simply points at the pills for 2–3 seconds; the system combines pill detection, schedule adherence, and visual proof in a single step, offering a practical middle ground between fully passive (impossible with current hardware) and fully manual systems.
- **Privacy-first continuous lifelogging implementation on constrained hardware**: Introduces automatic bystander face blurring, location-based recording pause, voice-activated privacy mode, and configurable data-expiry policies specifically engineered for the storage and bandwidth limitations of a camera + smartphone setup, setting a new benchmark for ethically acceptable lifelogging in real-

world conditions.

Caregiver dashboard with predictive behavioral alerts: Unlike existing elder-care monitors that only report location or falls, LOCUS provides early warnings for subtle but clinically significant changes (social withdrawal, disrupted sleep, routine abandonment) derived from long-term pattern analysis of first-person video, enabling earlier intervention.

Contribution Impact

LOCUS is different, as it demonstrates that effective cognitive aid does not mean having to wait until the future to use the futuristic glasses or spend thousands of dollars on bespoke hardware- a simple wearable camera can really benefit people in the present day. To a person with early dementia or some form of a brain injury who finds it difficult to recollect their routines in everyday life or who has suffered a brain injury and can no longer recall certain information, LOCUS silently observes their routines and interferes at the right time, such as prompting them to take their wallet or car keys and telling them where to last use it. This is critical to the 6 million Americans with impaired mind that cannot afford or operate complex assistive technology. Caregivers also benefit as opposed to being constantly on guard, they get early notifications when Mom has stopped going to her garden or the sleeping habits of Dad have changed- subtle cues that in most cases are indicative of the worsening of the situation well before a crisis has occurred. LOCUS seeks to solve the biggest barrier to people accepting lifelogging systems by accurately addressing the privacy issue with auto-blurred faces and an option to delete all data, which is the primary objection that people must lifelogging systems in the first place. Regarding research, the system changes the discussion not to what we will develop when the technology is more mature, but what we can implement today to assist people, and such practical orientation may transform how the whole field handles assistive AI. The anomaly-detection system and zero-configuration learning algorithm spawned by LOCUS are already finding applications in other areas besides cognitive assistance, researchers are accommodating mental health diagnosis, chronic illness management, and elder care. Finally, LOCUS illustrates how judicious application of existing tools can provide the sort of independence-saving support that was previously decades away, and that accomplishment will be what will probably jump start dozens of similar pragmatics first projects in the assistive technology field.

11. Relevance to Course Modules

LOCUS initiative combines fundamental computer science fields and directly employs principles from essential course modules:

11.1.1 Software Engineering

Applied comprehensive SDLC (Software Development Life Cycle) phases throughout the project:

- **Requirements Gathering**: User demands spanning groups (elderly depression assistance, general forgetfulness users) both functional and non-functional necessities, for memory retrieval routine acquisition and safety elements
- **Design Modeling**: Multi-layered system architecture with pattern recognition engines, behavior modeling components, and caregiver dashboards
- **Implementation**: Modular, scalable code structure with clear separation between data collection, AI processing, and user interface layers
- **Testing & Validation**: Ensuring quality, for anomaly detection methods, multi-modal memory retrieval capabilities and accessibility functionalities

11.1.2 Database Technologies

Structured data storage and retrieval systems:

- **Relational Database Design**: Normalized schemas for user profiles, medication schedules, social interactions, and location history
- **NoSQL Integration**: Document-based storage for unstructured memory data including images, audio recordings, and temporal event logs
- **Query Optimization**: Efficient indexing strategies for natural language memory queries and real-time pattern matching
- **Data Relationships**: Complex relationship mappings connecting objects, people, locations, and temporal data for cross-referenced memory access

11.1.3 Mobile App Development

Native mobile interfaces with Flutter for Android support:

- **Cross-Platform Development**: Unified codebase with responsive UI components
- **Real-Time Synchronization**: Live data sync between wearable device, mobile app, and caregiver dashboard
- **Offline Functionality**: Local storage and caching for critical features like medication reminders and emergency navigation
- **Push Notifications**: Escalating alert system with customizable notification priorities
- **Bluetooth Integration**: Communication protocols between wearable hardware and mobile application

11.1.4 Cybersecurity

Comprehensive security measures throughout the system:

- **Secure Authentication**: Multi-factor authentication with PIN/fingerprint biometric access
- **Data Encryption**: End-to-end encryption for all sensitive health data, location information, and personal memories
- **Privacy Protection**: Automatic face-blurring algorithms, configurable data expiry policies, and selective memory storage with user consent management
- **Access Control**: Role-based authorization system for caregiver dashboard with granular permission levels
- **Secure API Communication**: Protected communication channels with token-based authentication and rate limiting

12. Tools and Technologies

Table 2: Tools and Technologies for Proposed Project

	Tools	**Version**	**Rationale**
Tools And Technologies	Android Studio	2024.2.1	Mobile App Development IDE
	Visual Studio Code	1.95	Code Editor for Flutter Development
	Figma	Latest	UI/UX Design and Prototyping
	Postman	11.0	API Testing and Documentation
	Git	2.45	Version Control System
	Technology	**Version**	**Rationale**
	Flutter	3.24	Cross-Platform Mobile Development
	Dart	3.5	Programming Language for Flutter
	Firebase	11.0	Real-time Database and Authentication
	MongoDB	8.0	NoSQL Database for Memory Storage
	PostgreSQL	16.0	Relational Database Management
	Python	3.12	AI/ML Model Development
	TensorFlow Lite	2.15	On-Device Machine Learning
	Node.js	20.x	Backend API Development
	Express.js	4.19	Web Application Framework
	JWT	9.0	Secure Authentication Tokens
	Google Maps API	Latest	Location Services and Navigation
	OpenCV	4.9	Computer Vision for Face Recognition
	AWS S3	Latest	Cloud Storage for Media Files

13. Project Stakeholders and Roles

Table 3 Project Stakeholders for Proposed Project

Project Sponsor	COMSATS University Islamabad, Islamabad Campus
Stakeholder	• Muhammad Omair Tahir – Project Lead, Backend Development & AI/ML Integration • Taqdees Zahra – Mobile Application Development (iOS & Android), UI/UX Design & AI/ML Integration • Project Supervisor: Muhammad Rashid Mukhtar • Final Year Project Committee: Evaluation of project • End Users: Elderly individuals with memory concerns, Individuals with depression requiring routine support and General users experiencing forgetfulness

14. Module based Work Division

Table 4 Team Member Work Division for Proposed Project

Student Name	Student Registration Number	Responsibility/ Module / Feature
Muhammad Omair Tahir	SP23-BCS-059	**Module 1: Intelligent Routine Learning & Behavioral Analysis (FE-1 to FE-6)** • Implement activity monitoring and routine learning algorithms • Build behavioral profiles and temporal pattern analysis • Generate contextual alerts **Module 3: Face Recognition & Social Interaction Support (FE-1 to FE-5)** • Face recognition, matching, and interaction context retrieval **Module 5: Activity Summary & Monitoring Services (FE-1 to FE-5)** • Everyday summaries, behavioral insights, and storage monitoring

<table>
<tr>
<td></td>
<td></td>
<td>Module 6: Memory Search & Retrieval System (FE-1 to FE-6)
• Implement natural language processing- based text and voice memory search
• Develop filtering and visualization of search results
• Build daily activity timeline

Module 8: Caregiver Dashboard (FE-1 to FE-13)
• Live activity feed, maps, alerts, caregiver management
• Multi-channel alerts (email, SMS, push) with escalating protocols
• Reports, exports, and notification delivery tracking</td>
</tr>
<tr>
<td>Taqdees Zahra</td>
<td>SP23-BCS-096</td>
<td>Module 2: Multi-Modal Memory Capture & Processing (FE-1 to FE-6)
• Capture and process camera video/audio streams
• Implement CV, speech-to-text, privacy mode, and cloud sync

Module 4: Location Safety & Emergency Response (Elderly Users) (FE-1 to FE-5)
• GPS tracking and emergency triggers
• Voice-guided navigation and caregiver communication

Module 7: Medication Management & Health Tracking (FE-1 to FE-5)
• Medication scheduling and reminders
• Manual intake confirmation and snooze handling
• Caregiver notification logic for missed doses</td>
</tr>
</table>

		Module 9: User Dashboard (FE-1 to FE-12) • Personalized activity, task, and memory timeline dashboard • Medication status, alerts, and notification management • System controls, device health, storage monitoring, and notification scheduling

Shared Responsibilities (All Team Members):

Integration of Camera, Mobile, and Cloud (Collaborative)

- **Omair:** Develops the API for the camera to communicate raw data to the processing unit and design the RESTful API endpoints on the server.
- **Taqdees:** Implements the Bluetooth/Wi-Fi bridging in the Mobile App to receive processed alerts from Omair's module and receive synced data from the Mobile App.

Data Privacy & Security (Collaborative)

- **Omair:** Implements local encryption for video files on the camera/device storage and ensures encryption for data at rest in the cloud database and manages role-based access control.
- **Taqdees:** Implements secure login and session management on the mobile application.

Testing & Quality Assurance (All Members)

- **Omair:** Unit testing for Machine Learning models and computer vision accuracy and load testing for server capabilities, notification delivery latency, and database integrity.
- **Taqdees:** User Acceptance Testing (UAT) for mobile UI/UX and elderly accessibility standards.

Project Management

- Weekly code reviews and integration testing sessions.
- Joint creation of final technical documentation and user manuals.

15. WBS and Gantt Chart

WBS and Gantt Chart:

Level 1	Level 2	Level 3
Camera Application		
	1.1 Module 1: Intelligent Routine Learning & Behavioral Analysis	
		1.1.1 FE-1: Monitor User Activities
		1.1.2 FE-2: Learn User Schedules & Behaviors
		1.1.3 FE-3: Build Behavioral Profiles
		1.1.4 FE-4: Track Temporal Patterns
		1.1.5 FE-5: Generate Contextual Alerts
	1.2 Module 2: Multi-Modal Memory Capture & Processing	
		1.2.1 FE-1: Capture Video & Audio Feeds
		1.2.2 FE-2: Process Visual Data with Computer Vision
		1.2.3 FE-3: Extract Audio & Speech-to-Text
		1.2.4 FE-4: Store Patterns with Cloud Sync
		1.2.5 FE-5: Enable Privacy Mode
		1.2.6 FE-6: Support Location-Based Exclusion

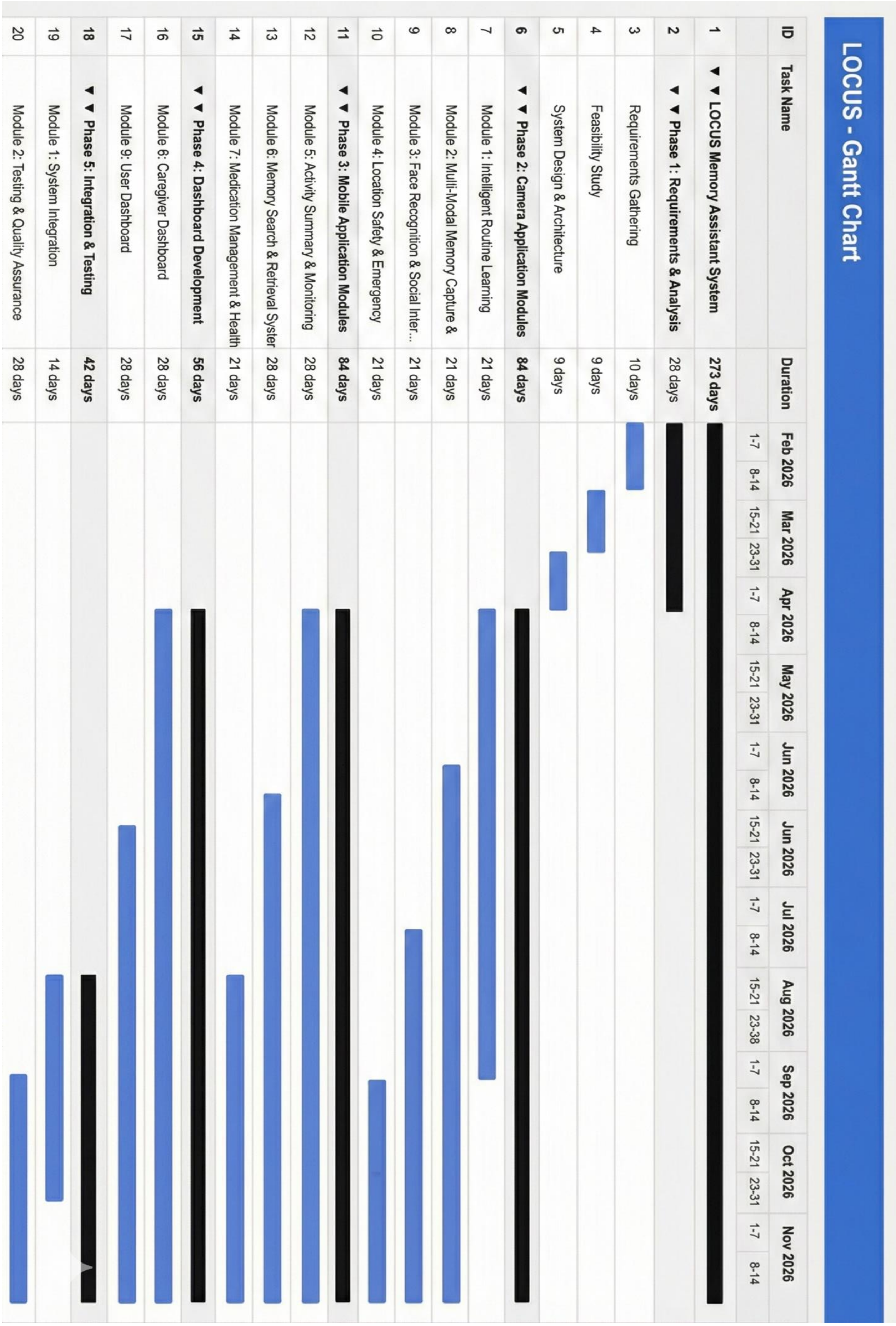
	1.3 Module 3: Face Recognition & Social Interaction Support	
		1.3.1 FE-1: Real-Time Face Detection
		1.3.2 FE-2: Match Faces Against Database
		1.3.3 FE-3: Provide Contextual Information
		1.3.4 FE-4: Store New Face Data
		1.3.5 FE-5: Display People Reminders
	1.4 Module 4: Location Safety & Emergency Response (Elderly Users)	
		1.4.1 FE-1: Continuous GPS Tracking
		1.4.2 FE-2: Trigger Emergency Mode
		1.4.3 FE-3: Provide Turn-by-Turn Navigation
		1.4.4 FE-4: Activate Location Sharing
		1.4.5 FE-5: Enable Two-Way Communication
Mobile Application (iOS & Android)		
	2.1 Module 5: Activity Summary & Monitoring Services	
		2.1.1 FE-1: Display Daily Activity Summary
		2.1.2 FE-2: Show Medication Adherence Status
		2.1.3 FE-3: Present Behavioral Insights
		2.1.4 FE-4: Provide Quick Access to Memories
		2.1.5 FE-5: Monitor Storage Usage
	2.2 Module 6: Memory Search & Retrieval System	
		2.2.1 FE-1: Enable Natural Language Search
		2.2.2 FE-2: Support Voice Input
		2.2.3 FE-3: Filter Search Results
		2.2.4 FE-4: Display Search Results with Metadata
		2.2.5 FE-5: Combine Multi-Modal Search

		2.2.6 FE-6: Provide Timeline Visualization
	2.3 Module 7: Medication Management & Health Tracking	
		2.3.1 FE-1: View Medication Schedule
		2.3.2 FE-2: Receive Push Notifications
		2.3.3 FE-3: Mark Doses Manually
		2.3.4 FE-4: Set Snooze Options
		2.3.5 FE-5: View Caregiver Notifications
Dashboard Modules		
	3.1 Module 8: Caregiver Dashboard - Web & Mobile	
		3.1.1 FE-1: View Live Activity Feed
		3.1.2 FE-2: Display Real-Time Location Map
		3.1.5 FE-3: Access Emergency Location Data
		3.1.6 FE-4: Send Custom Reminders
		3.1.7 FE-5: Request Status Check
		3.1.10 FE-6: Manage Caregiver Accounts
		3.1.11 FE-7: Configure Notification Preferences
		3.1.12 FE-8: Manage Caregiver Profiles
		3.1.15 FE-9: Download Location Data
		3.1.10 FE-10: Send Email Notifications
		3.1.11 FE-11: Deliver SMS Alerts
		3.1.12 FE-12: Implement Escalating Alert
		3.1.13 FE-13: Track Notification Delivery Status

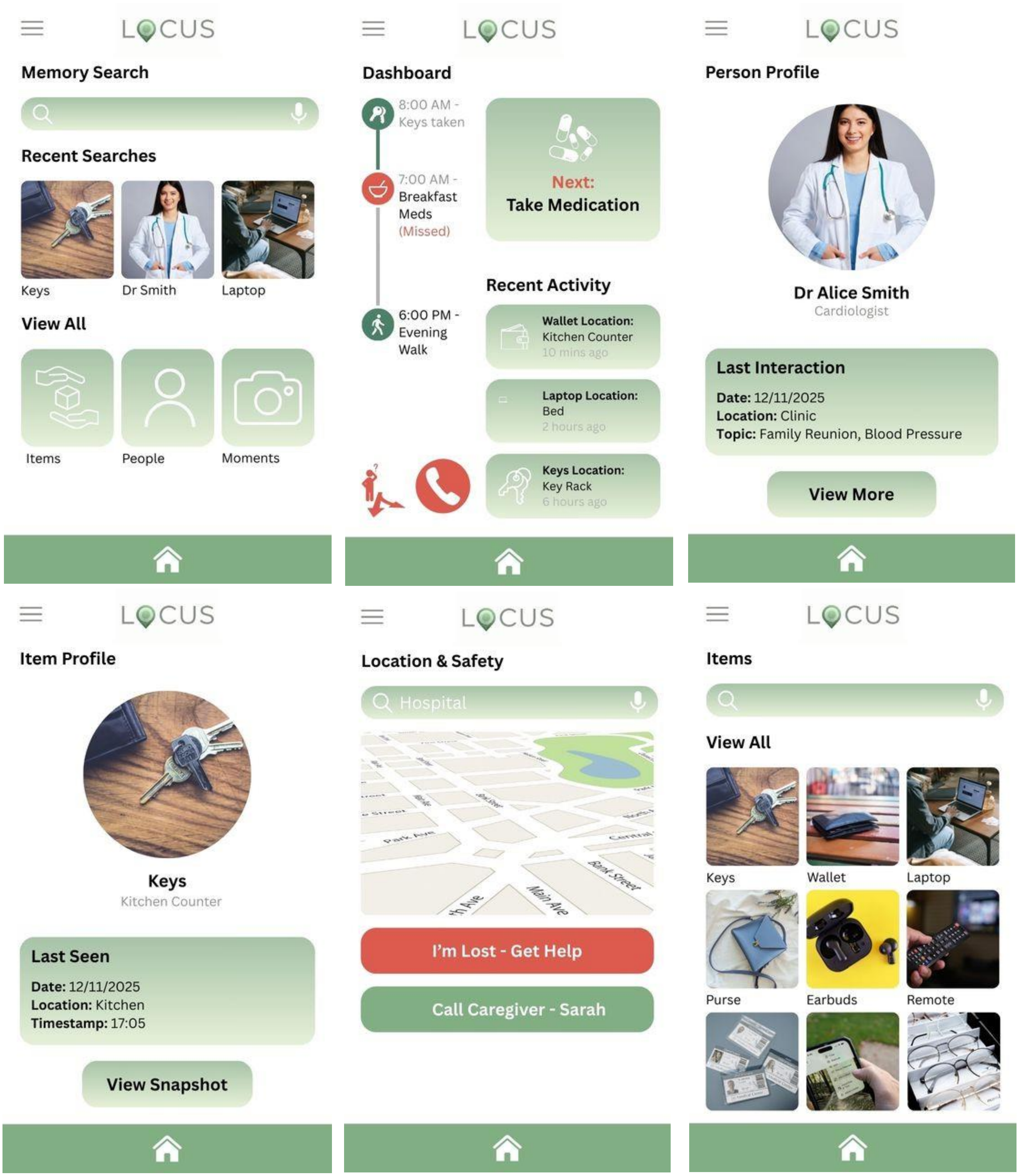
	3.2 Module 9: User Dashboard – Web & Mobile (Normal Users)	
		3.2.1 FE-1: Show Personalized Daily Summary
		3.2.3 FE-2: Display Smart To-Do Progress
		3.2.6 FE-3: Display Face Recognition Insights
		3.2.7 FE-4: Show Medication Schedule & Status
		3.2.8 FE-5: Show Alert History
		3.2.9 FE-6: Allow Alert Management
		3.2.11 FE-7: Provide One-Tap Actions
		3.2.12 FE-8: Show Location History
		3.2.14 FE-9: Display Storage Usage
		3.2.10 FE-10: Manage Notification Queue
		3.2.11 FE-11: Handle Push Notifications
		3.2.12 FE-12: Support Notification Scheduling

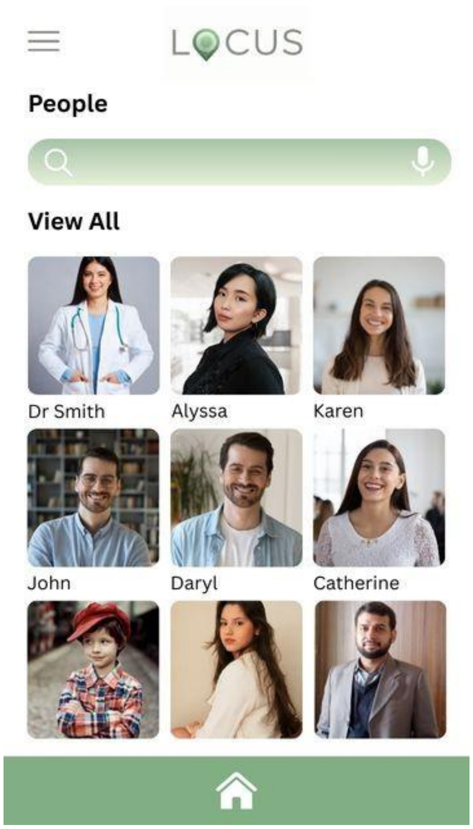
Table A-1 – WBS for LOCUS

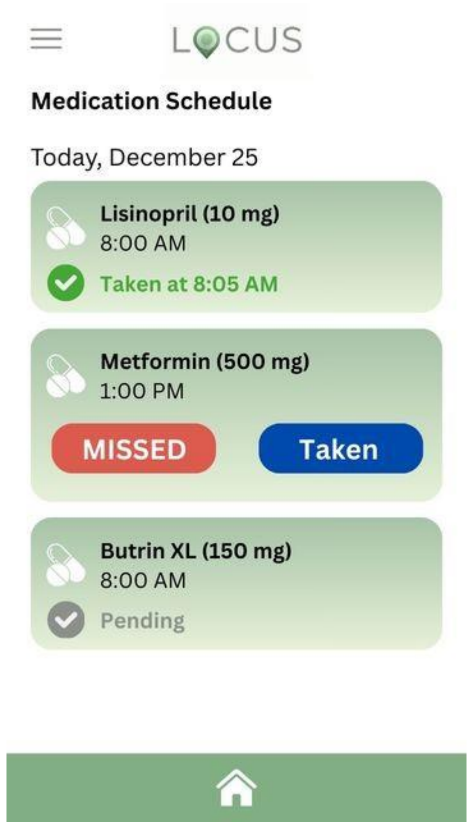
ID		Task	Duration	Resources
1	**Analysis**		**42d**	**Omair; Taqdees**
2		Requirement Meetings	14d	Omair; Taqdees
3		Communication with Stakeholders	14 d	Taqdees
4		Document Current System	14d	Omair
5		Analysis Finished	1d	
6	**Design**		**56d**	**Omair; Taqdees**
7		Design Database	21d	Omair; Taqdees
8		Software Design	21d	Taqdees
9		Interface Design	14d	Omair; Taqdees
10		Create Design Specifications	21d	Omair
11		Design Finished	1d	
12	**Development**		**84d**	**Omair; Taqdees**
13		Develop System Module	42d	Omair; Taqdees
14		Integrate System Module	28d	Omair
15		Perform Initial Testing	14d	Taqdees
16		Development Finished	1d	
17	**Testing**		**56d**	**Omair; Taqdees**
18		Perform System Testing	21d	Omair
19		Document Issues Found	14d	Taqdees
20		Correct Issues Found	21d	Omair
21		Testing Finished	1d	
22	**Implementation**		**28d**	**Omair; Taqdees**
23		On-Site Installation	14d	Omair
24		Support Plan for the System	14d	Taqdees
25	**Completion**		**42d**	**Omair; Taqdees**
26		System Maintenance	21d	Taqdees
27		Evaluation	21d	Omair

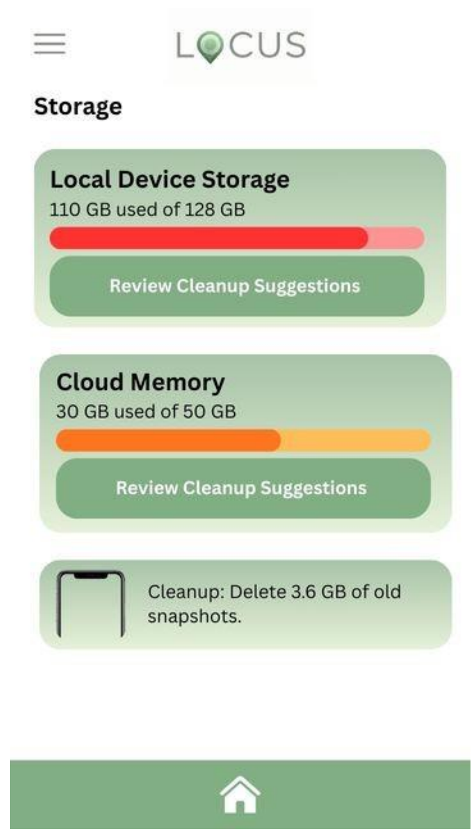

LOCUS - Gantt Chart
ID
Task Name
Duration
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jun 2026
Jul 2026
Aug 2026
Sep 2026
Oct 2026
Nov 2026
1-7
8-14
15-21
23-31
1-7
8-14
15-21
23-31
1-7
8-14
15-21
23-31
1-7
8-14
15-21
23-38
1-7
8-14
15-21
23-31
1-7
8-14
1
▼ ▼ LOCUS Memory Assistant System
273 days
2
▼ ▼ Phase 1: Requirements & Analysis
28 days
3
Requirements Gathering
10 days
4
Feasibility Study
9 days
5
System Design & Architecture
9 days
6
▼ ▼ Phase 2: Camera Application Modules
84 days
7
Module 1: Intelligent Routine Learning
21 days
8
Module 2: Multi-Modal Memory Capture &
21 days
9
Module 3: Face Recognition & Social Inter...
21 days
10
Module 4: Location Safety & Emergency
21 days
11
▼ ▼ Phase 3: Mobile Application Modules
84 days
12
Module 5: Activity Summary & Monitoring
28 days
13
Module 6: Memory Search & Retrieval Syster
28 days
14
Module 7: Medication Management & Health
21 days
15
▼ ▼ Phase 4: Dashboard Development
56 days
16
Module 6: Caregiver Dashboard
28 days
17
Module 9: User Dashboard
28 days
18
▼ ▼ Phase 5: Integration & Testing
42 days
19
Module 1: System Integration
14 days
20
Module 2: Testing & Quality Assurance
28 days

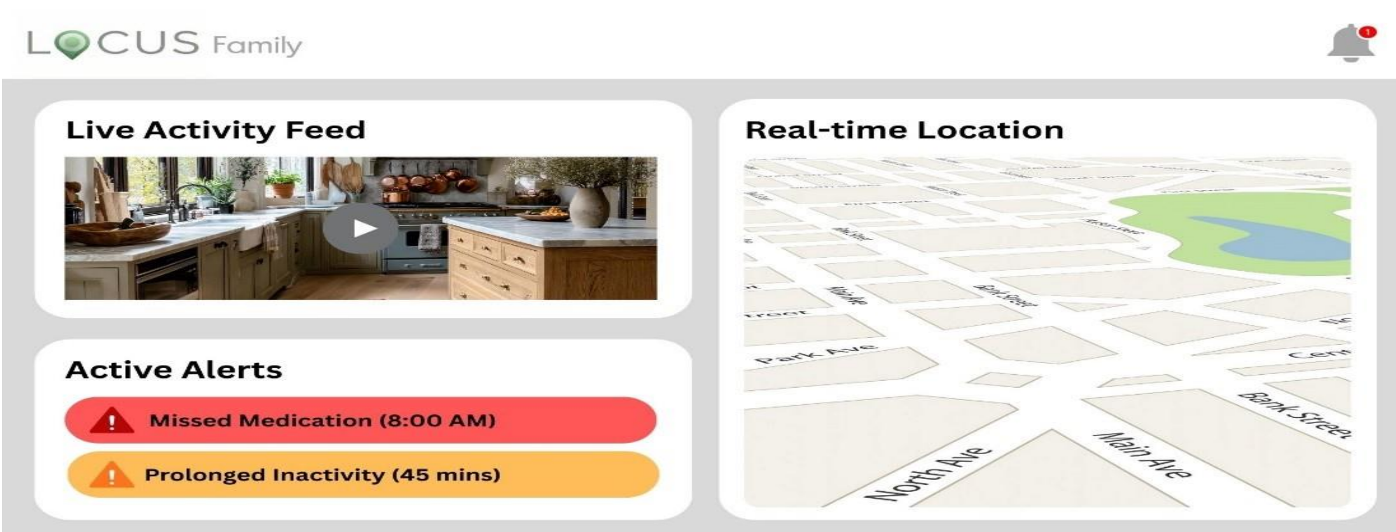
16. Mockups


LOCUS
Memory Search
Recent Searches
Keys
Dr Smith
Laptop
View All
Items
People
Moments
LOCUS
Dashboard
8:00 AM - Keys taken
7:00 AM - Breakfast Meds (Missed)
6:00 PM - Evening Walk
Next:
Take Medication
Recent Activity
Wallet Location: Kitchen Counter
10 mins ago
Laptop Location: Bed
2 hours ago
Keys Location: Key Rack
6 hours ago
LOCUS
Person Profile
Dr Alice Smith
Cardiologist
Last Interaction
Date: 12/11/2025
Location: Clinic
Topic: Family Reunion, Blood Pressure
View More
LOCUS
Item Profile
Keys
Kitchen Counter
Last Seen
Date: 12/11/2025
Location: Kitchen
Timestamp: 17:05
View Snapshot
LOCUS
Location & Safety
Hospital
Park Ave
Main Ave
Bank Street
I'm Lost - Get Help
Call Caregiver - Sarah
LOCUS
Items
View All
Keys
Wallet
Laptop
Purse
Earbuds
Remote


LOCUS
People
View All
Dr Smith
Alyssa
Karen
John
Daryl
Catherine


LOCUS
Medication Schedule
Today, December 25
Lisinopril (10 mg)
8:00 AM
Taken at 8:05 AM
Metformin (500 mg)
1:00 PM
MISSED
Taken
Butrin XL (150 mg)
8:00 AM
Pending


LOCUS
Storage
Local Device Storage
110 GB used of 128 GB
Review Cleanup Suggestions
Cloud Memory
30 GB used of 50 GB
Review Cleanup Suggestions
Cleanup: Delete 3.6 GB of old snapshots.


LOCUS Family
Live Activity Feed
Active Alerts
Missed Medication (8:00 AM)
Prolonged Inactivity (45 mins)
Real-time Location
Park Ave
North Ave
Main Ave
Bank Street

1. References

Li, X., et al. "FMT: Fiducial Marker Tracker for Object Tracking with Neck-Mounted Cameras." *Proc. International Conference on Computer Vision (ICCV)*, 2019, pp. 123-130.

Sato, Y. "GO-Finder: Real-Time Object Retrieval from Continuous Video Streams." *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 4578-4587.

Hodges, S., et al. "SenseCam: A Retrospective Memory Aid." *UbiComp 2006: Ubiquitous Computing Conference Proceedings*, 2006, pp. 177-193.

Maniar, A., et al. "MemPal: Multimodal AI Assistant for Wearable Memory Support Using Vision-Language Models." *Journal of Ambient Intelligence and Humanized Computing*, vol. 15, pp. 2345-2362, 2024.

O'Brien, K., et al. "Voice-Controlled Intelligent Personal Assistants for Older Adults: Use and Perceptions." *International Journal of Human-Computer Interaction*, vol. 38, pp. 1102-1118, 2022.

Tsiourti, C., et al. "Assistive Companions for Elderly Adults: Field Study on Acceptance and Design Factors." *Journal of Assistive Technologies*, vol. 8, pp. 45-60, 2014.

2. Plagiarism Report


turnitin Page 2 of 38 - Integrity Overview Submission ID trn:oid:::3618:124466121

5% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography

Match Groups

- 25 Not Cited or Quoted 5% — Matches with neither in-text citation nor quotation marks
- 1 Missing Quotations 0% — Matches that are still very similar to source material
- 0 Missing Citation 0% — Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0% — Matches with in-text citation present, but no quotation marks

Top Sources

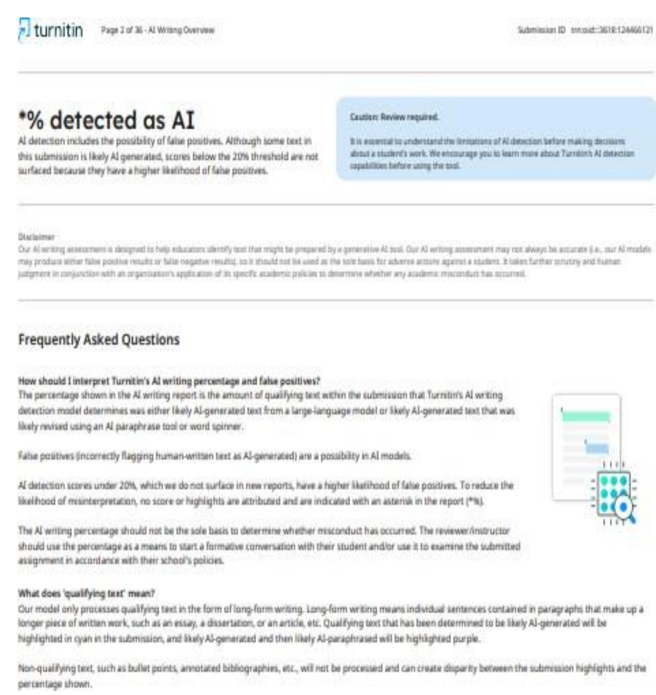
- 3% Internet sources
- 0% Publications
- 5% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.


turnitin Page 2 of 36 - AI Writing Overview Submission ID trn:oid:::3618:124466121

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Final Year Project (FYP) – Supervisor Approval Form

Student Information

- **Student Name:** Muhammad Omair Tahir, Taqdees Zahra
- **Registration Number:** SP23-BCS-059, SP23-BCS-096
- **Program of Study:** Computer Science
- **Department:** Computer Science
- **University/Campus:** Comsats University Islamabad

Supervisor Information

- **Supervisor Name:** Muhammad Rashid Mukhtar
- **Designation:** Assistant Professor
- **Department:** Computer Science
- **Email Address:** rashid.mukhtar@comsats.edu.pk
- **Contact Number:** 05190495366

FYP Details

- **Title of the FYP (if finalized):**

  LOCUS

- **Area of Research:**

  Assistive Technology & Wearable Computing

Project Description

Aim of the Project

(Briefly state the overall aim of the project)

To create an integrated, AI-driven platform that provides continuous context-aware monitoring and personalized cognitive support for individuals with memory impairments, seniors, and busy professionals.

Objectives

(List the specific objectives of the project)

BO-1: Achieve 80-90% accuracy in automated learning and identification of behavior patterns.

BO-2: Decrease medication non-adherence by 60% through clever checks and caregiver integration

BO-3: Reduce time spent locating lost items by 70% via multi-modal memory retrieval.

BO-4: Cut wandering-related accidents by 50% among elderly users through proactive anomaly detection.

Methodology

(Briefly describe the methodology/approach to be used in the project)

A Wearable Camera will be used for continuous monitoring and data capture with a cloud-based AI to handle video processing, VLM-powered memory indexing, and anomaly detection and a Mobile/Web Dashboard for real-time activity summaries, medication tracking, and caregiver alerts.

Expected Contributions / Outcomes

(Describe the expected contributions)

For Users: Enhanced independence and quality of life through memory-on-demand and proactive routine assistance.

For Caregivers: Reduced emotional and physical pressure through remote monitoring and predictive behavioral alerts.

Supervisor Declaration

I hereby declare that: - I have carefully read and reviewed the **FYP scope document** submitted by the student. - I have also checked the **attached project presentation**. - I am satisfied with the clarity, feasibility, and relevance of the proposed Final Year Project. - I assure that the proposed project is appropriate for the level of the program and can be completed within the allocated time under my supervision.

I agree to supervise the above-mentioned student for the proposed Final Year Project.

Signatures

- **Supervisor Signature:** ________________________________ **Date:** 15/01/2026
- **Student Signature:** ________________________________ **Date:** 15/01/2026